

Supplementary Table 2: All *BAP1* missense variants associated with KURIS/NDD and TPDS, and their ClinVar B/LB controls

KURIS/NDD missense variants, TPDS-associated ClinVar pathogenic/likely-pathogenic missense variants, and ClinVar benign/likely-benign controls, with HGVS nomenclature, domain, SGE functional scores and class, and ClinVar classifications. The SGE evidence column gives the ACMG code (PS3 Strong / BS3 Moderate) from the KURIS/NDD-vs-B/LB calibration (Supplementary Table 4) for the KURIS/NDD and B/LB rows; it is blank for the TPDS P/LP rows, which are not part of that comparison.

Variant Category	HGVS g.	HGVS c.	HGVS p.	Domain	SGE score	SGE class	SGE evidence	ClinVar (agg. germline)	ClinVar VCV	Stars
KURIS/NDD (proband)	NC_000003.12:g.52406349G>T	NM_004656.4:c.687C>A	NP_004647.1:p.Asn229Lys	UCH	-0.200	Abnormal	PS3 Strong	Likely pathogenic	VCV002429331	2
KURIS/NDD	NC_000003.12:g.52409585C>T	NM_004656.4:c.91G>A	NP_004647.1:p.Glu31Lys	UCH	-0.201	Abnormal	PS3 Strong	Conflicting classifications of pathogenicity	VCV001173078	1
KURIS/NDD	NC_000003.12:g.52408061C>G	NM_004656.4:c.272G>C	NP_004647.1:p.Cys91Ser	UCH	-0.177	Abnormal	PS3 Strong	Pathogenic	VCV001173081	1
KURIS/NDD	NC_000003.12:g.52407247G>C	NM_004656.4:c.507C>G	NP_004647.1:p.His169Gln	UCH	-0.169	Abnormal	PS3 Strong	Likely pathogenic	VCV003132975	1
KURIS/NDD	NC_000003.12:g.52409851T>G	NM_004656.4:c.28A>C	NP_004647.1:p.Ser10Arg	UCH	-0.162	Abnormal	PS3 Strong			
KURIS/NDD	NC_000003.12:g.52408062A>C	NM_004656.4:c.271T>G	NP_004647.1:p.Cys91Gly	UCH	-0.161	Abnormal	PS3 Strong	Uncertain significance	VCV001173082	1
KURIS/NDD	NC_000003.12:g.52407248T>C	NM_004656.4:c.506A>G	NP_004647.1:p.His169Arg	UCH	-0.137	Abnormal	PS3 Strong	Conflicting classifications of pathogenicity	VCV001173083	1
KURIS/NDD	NC_000003.12:g.52409845G>T	NM_004656.4:c.34C>A	NP_004647.1:p.Pro12Thr	UCH	-0.137	Abnormal	PS3 Strong	Pathogenic	VCV001064792	0
KURIS/NDD	NC_000003.12:g.52408062A>G	NM_004656.4:c.271T>C	NP_004647.1:p.Cys91Arg	UCH	-0.135	Abnormal	PS3 Strong	Conflicting classifications of pathogenicity	VCV000988095	1
KURIS/NDD	NC_000003.12:g.52408583A>G	NM_004656.4:c.146T>C	NP_004647.1:p.Leu49Pro	UCH	-0.135	Abnormal	PS3 Strong	Likely pathogenic	VCV001173079	1
KURIS/NDD	NC_000003.12:g.52408553C>G	NM_004656.4:c.176G>C	NP_004647.1:p.Arg59Pro	UCH	-0.102	Abnormal	PS3 Strong	Pathogenic	VCV001173080	1
KURIS/NDD	NC_000003.12:g.52409845G>C	NM_004656.4:c.34C>G	NP_004647.1:p.Pro12Ala	UCH	-0.078	Abnormal	PS3 Strong	Likely pathogenic	VCV001173077	1
KURIS/NDD	NC_000003.12:g.52409578T>G	NM_004656.4:c.98A>C	NP_004647.1:p.Tyr33Ser	UCH	-0.072	Abnormal	PS3 Strong			
KURIS/NDD	NC_000003.12:g.52408058G>A	NM_004656.4:c.275C>T	NP_004647.1:p.Ala92Val	UCH	-0.062	Abnormal	PS3 Strong	Pathogenic/Likely pathogenic	VCV002681785	2
KURIS/NDD	NC_000003.12:g.52409851T>C	NM_004656.4:c.28A>G	NP_004647.1:p.Ser10Gly	UCH	-0.056	Abnormal	PS3 Strong	Likely pathogenic	VCV003377067	1
KURIS/NDD	NC_000003.12:g.52408550C>T	NM_004656.4:c.179G>A	NP_004647.1:p.Arg60Gln	UCH	-0.018	Normal	BS3 Moderate	Conflicting classifications of pathogenicity	VCV000489626	1
KURIS/NDD	NC_000003.12:g.52402325C>T	NM_004656.4:c.2153G>A	NP_004647.1:p.Arg718Gln	BRCA1-binding	0.003	Normal	BS3 Moderate	Uncertain significance	VCV001173084	2
KURIS/NDD	NC_000003.12:g.52408553C>T	NM_004656.4:c.176G>A	NP_004647.1:p.Arg59Gln	UCH	0.007	Normal	BS3 Moderate	Conflicting classifications of pathogenicity	VCV001320722	1
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52409848C>G	NM_004656.4:c.31G>C	NP_004647.1:p.Asp11His	UCH	-0.217	Abnormal		Likely pathogenic	VCV000823162	1
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52406884A>G	NM_004656.4:c.604T>C	NP_004647.1:p.Trp202Arg	UCH	-0.178	Abnormal		Pathogenic/Likely pathogenic	VCV001399300	2
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52406882C>G	NM_004656.4:c.606G>C	NP_004647.1:p.Trp202Cys	UCH	-0.175	Abnormal		Likely pathogenic	VCV002450417	1
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52408596C>T	NM_004656.4:c.133G>A	NP_004647.1:p.Gly45Arg	UCH	-0.153	Abnormal		Likely pathogenic	VCV000485289	2
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407958C>G	NM_004656.4:c.375G>C	NP_004647.1:p.Glu125Asp	UCH	-0.146	Abnormal		Likely pathogenic	VCV003477532	1
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407399C>G	NM_004656.4:c.437G>C	NP_004647.1:p.Arg146Thr	UCH	-0.144	Abnormal		Likely pathogenic	VCV001698836	2
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52402643T>C	NM_004656.4:c.2015A>G	NP_004647.1:p.Asp672Gly	ULD	-0.140	Abnormal		Likely pathogenic	VCV002626728	1
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407415G>A	NM_004656.4:c.421C>T	NP_004647.1:p.His141Tyr	UCH	-0.140	Abnormal		Likely pathogenic	VCV001738816	1

Variant Category	HGVS g.	HGVS c.	HGVS p.	Domain	SGE score	SGE class	SGE evidence	ClinVar (agg. germline)	ClinVar VCV	Stars
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407454C>T	NM_004656.4:c.382G>A	NP_004647.1:p.Gly128Arg	UCH	-0.136	Abnormal		Pathogenic	VCV004075813	1
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52403437G>C	NM_004656.4:c.1708C>G	NP_004647.1:p.Leu570Val	Other	-0.122	Abnormal		Pathogenic/Likely pathogenic	VCV001778494	2
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407215A>G	NM_004656.4:c.539T>C	NP_004647.1:p.Leu180Pro	UCH	-0.119	Abnormal		Likely pathogenic	VCV003477629	1
Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52406902A>G	NM_004656.4:c.586T>C	NP_004647.1:p.Trp196Arg	UCH	-0.116	Abnormal		Likely pathogenic	VCV004294247	1
ClinVar B/LB (control)	NC_000003.12:g.52403199C>T	NM_004656.4:c.1829G>A	NP_004647.1:p.Arg610Lys	BRCA1-binding	-0.016	Normal	BS3 Moderate	Likely benign	VCV003224444	2
ClinVar B/LB (control)	NC_000003.12:g.52403242T>C	NM_004656.4:c.1786A>G	NP_004647.1:p.Ser596Gly	BRCA1-binding	-0.007	Normal	BS3 Moderate	Benign	VCV000133664	2
ClinVar B/LB (control)	NC_000003.12:g.52405827T>G	NM_004656.4:c.869A>C	NP_004647.1:p.Asn290Thr	Other	-0.005	Normal	BS3 Moderate	Likely benign	VCV003477925	1
ClinVar B/LB (control)	NC_000003.12:g.52405279G>T	NM_004656.4:c.947C>A	NP_004647.1:p.Ala316Glu	Other	-0.004	Normal	BS3 Moderate	Likely benign	VCV002565171	1
ClinVar B/LB (control)	NC_000003.12:g.52404485C>G	NM_004656.4:c.1218G>C	NP_004647.1:p.Glu406Asp	Other	-0.002	Normal	BS3 Moderate	Likely benign	VCV001752271	1
ClinVar B/LB (control)	NC_000003.12:g.52403274G>A	NM_004656.4:c.1754C>T	NP_004647.1:p.Ser585Phe	Other	-0.002	Normal	BS3 Moderate	Likely benign	VCV003477483	1
ClinVar B/LB (control)	NC_000003.12:g.52403292C>T	NM_004656.4:c.1736G>A	NP_004647.1:p.Gly579Glu	Other	-0.001	Normal	BS3 Moderate	Benign	VCV002450425	1
ClinVar B/LB (control)	NC_000003.12:g.52403718A>G	NM_004656.4:c.1427T>C	NP_004647.1:p.Val476Ala	Other	-0.001	Normal	BS3 Moderate	Benign/Likely benign	VCV000240048	2
ClinVar B/LB (control)	NC_000003.12:g.52402827G>C	NM_004656.4:c.1935C>G	NP_004647.1:p.Asn645Lys	BRCA1-binding	-0.001	Normal	BS3 Moderate	Benign	VCV004194475	1
ClinVar B/LB (control)	NC_000003.12:g.52405812A>G	NM_004656.4:c.884T>C	NP_004647.1:p.Val295Ala	Other	-0.001	Normal	BS3 Moderate	Benign	VCV003477649	1
ClinVar B/LB (control)	NC_000003.12:g.52404549C>G	NM_004656.4:c.1154G>C	NP_004647.1:p.Arg385Pro	Other	-0.001	Normal	BS3 Moderate	Likely benign	VCV000818436	1
ClinVar B/LB (control)	NC_000003.12:g.52403235G>A	NM_004656.4:c.1793C>T	NP_004647.1:p.Pro598Leu	BRCA1-binding	0.001	Normal	BS3 Moderate	Benign	VCV003818386	1
ClinVar B/LB (control)	NC_000003.12:g.52405890T>C	NM_004656.4:c.139A>G	NP_004647.1:p.Ile47Val	UCH	0.001	Normal	BS3 Moderate	Benign	VCV001771715	1
ClinVar B/LB (control)	NC_000003.12:g.52403877G>T	NM_004656.4:c.1268C>A	NP_004647.1:p.Thr423Lys	Other	0.002	Normal	BS3 Moderate	Benign/Likely benign	VCV000240045	2
ClinVar B/LB (control)	NC_000003.12:g.52402828T>C	NM_004656.4:c.1934A>G	NP_004647.1:p.Asn645Ser	BRCA1-binding	0.003	Normal	BS3 Moderate	Likely benign	VCV003260366	1
ClinVar B/LB (control)	NC_000003.12:g.52403162C>G	NM_004656.4:c.1866G>C	NP_004647.1:p.Leu622Phe	BRCA1-binding	0.003	Normal	BS3 Moderate	Benign	VCV004620225	1
ClinVar B/LB (control)	NC_000003.12:g.52405795C>A	NM_004656.4:c.901G>T	NP_004647.1:p.Ala301Ser	Other	0.004	Normal	BS3 Moderate	Benign	VCV004620599	1
ClinVar B/LB (control)	NC_000003.12:g.52407418C>T	NM_004656.4:c.418G>A	NP_004647.1:p.Ala140Thr	UCH	0.004	Normal	BS3 Moderate	Likely benign	VCV004194127	1
ClinVar B/LB (control)	NC_000003.12:g.52405791G>A	NM_004656.4:c.905C>T	NP_004647.1:p.Pro302Leu	Other	0.004	Normal	BS3 Moderate	Benign/Likely benign	VCV000133661	2
ClinVar B/LB (control)	NC_000003.12:g.52405827T>C	NM_004656.4:c.869A>G	NP_004647.1:p.Asn290Ser	Other	0.005	Normal	BS3 Moderate	Benign/Likely benign	VCV000472716	2
ClinVar B/LB (control)	NC_000003.12:g.52408493T>C	NM_004656.4:c.236A>G	NP_004647.1:p.Asn79Ser	UCH	0.005	Normal	BS3 Moderate	Likely benign	VCV003224460	1
ClinVar B/LB (control)	NC_000003.12:g.52409555C>T	NM_004656.4:c.121G>A	NP_004647.1:p.Gly41Ser	UCH	0.006	Normal	BS3 Moderate	Benign/Likely benign	VCV000133667	2
ClinVar B/LB (control)	NC_000003.12:g.52404520A>C	NM_004656.4:c.1183T>G	NP_004647.1:p.Ser395Ala	Other	0.007	Normal	BS3 Moderate	Benign	VCV003986661	1
ClinVar B/LB (control)	NC_000003.12:g.52403190G>A	NM_004656.4:c.1838C>T	NP_004647.1:p.Thr613Met	BRCA1-binding	0.009	Normal	BS3 Moderate	Benign/Likely benign	VCV000133665	2
ClinVar B/LB (control)	NC_000003.12:g.52407990T>G	NM_004656.4:c.343A>C	NP_004647.1:p.Met115Leu	UCH	0.014	Normal	BS3 Moderate	Benign	VCV004620701	1
ClinVar B/LB (control)	NC_000003.12:g.52403646C>T	NM_004656.4:c.1499G>A	NP_004647.1:p.Gly500Asp	Other	0.014	Normal	BS3 Moderate	Likely benign	VCV004620450	1
ClinVar B/LB (control)	NC_000003.12:g.52409574G>C	NM_004656.4:c.102C>G	NP_004647.1:p.Asp34Glu	UCH	0.014	Normal	BS3 Moderate	Likely benign	VCV003986856	1
ClinVar B/LB (control)	NC_000003.12:g.52403799G>A	NM_004656.4:c.1346C>T	NP_004647.1:p.Ala449Val	Other	0.016	Normal	BS3 Moderate	Benign	VCV003477759	1